

Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024: A Source-Audited Panel Analysis

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Abstract

Permitless concealed carry laws allow eligible adults to carry concealed handguns in public without first obtaining a carry permit. The rapid spread of these laws creates a timely policy question, but estimation is difficult because adoption is politically patterned, legal classifications contain edge cases, and permitless-carry statutes often remove several pre-carry requirements at once. We assembled a 50-state annual panel from 1999 through 2024 using CDC mortality outcomes and a source-audited permitless-carry treatment file. Firearm suicide was the prespecified primary outcome; total suicide, non-firearm suicide, firearm homicide, and total firearm deaths were secondary outcome-family checks. The primary estimand compares 26 source-verified within-panel adopters with 21 verified non-adopters; ambiguous, partial, and baseline-permitless states are handled outside the primary model. In the baseline fixed-effects model with state and year fixed effects, unemployment, income, and state-clustered standard errors, permitless carry adoption was associated with a 1.391 deaths per 100,000 increase in firearm suicide rates (SE 0.222; $p < 0.001$). Age-adjusted firearm-suicide models gave a similar estimate (1.407; SE 0.228; $p < 0.001$). The estimate remained positive in population-weighted models (0.919; $p < 0.001$), pre-COVID and COVID-excluded samples, and staggered-adoption summaries based on cohort windows and not-yet-treated comparisons. It attenuated with state-specific linear trends (0.249; $p = 0.079$), and event-time estimates showed statistically significant pre-adoption signal. Firearm homicide showed no comparable baseline association (−0.234; SE 0.338; $p = 0.489$), although homicide-rate missingness was concentrated in the smallest-population state-years. Overall, permitless carry adoption was most consistently associated with firearm suicide and total suicide, not firearm homicide. The findings support a state-level observational association and should not be interpreted as individual-level causal evidence.

Introduction

Permitless concealed carry is a newer policy margin in a long-running concealed-carry debate. Earlier right-to-carry studies focused mostly on transitions from no-issue or may-issue systems to shall-issue permitting, asking whether broader legal carrying deters violent crime or increases firearm violence [1–3]. That literature remains methodologically contested, with replication work emphasizing sensitivity to event-study validity, heterogeneous treatment effects, and influential observations [4]. The policy landscape has since shifted again. Since 2015, many states have moved beyond shall-issue permitting by allowing eligible adults to carry concealed handguns without

first obtaining a carry permit, a change that became still more salient after *Bruen* altered the constitutional environment for restrictive carry permitting [5]. The relevant empirical question is no longer only whether permits should be issued more easily, but what happens when the permit requirement is removed.

That distinction is substantively important. A carry-permit requirement can screen applicants, impose delay, require training, or document who is legally authorized to carry. A permitless-carry statute may remove a training requirement, a permit-background-check step, a permit-specific misdemeanor screen, or a discretionary suitability review, while leaving general firearm-possession prohibitions in place. Prior concealed-carry work has emphasized that these permit provisions can matter for firearm violence outcomes [2]. A recent permitless-carry working paper estimates increases in violent crime and firearm aggravated assault after adoption, reinforcing the need to evaluate permitless carry separately from older shall-issue transitions [6]. Recent firearm-suicide work has also reported that concealed-carry permit requirements are associated with lower firearm suicide rates, alongside handgun purchase permits and waiting periods [7]. The policy margin is therefore relevant to both suicide and homicide, but the pathway is not obvious *ex ante*.

Suicide and homicide also require separate analysis. Firearm access is strongly associated with firearm suicide at household, individual, and state levels [8–10]. Purchaser-licensing evidence links stronger handgun screening systems with lower firearm homicide and suicide in several state settings [11]. A law governing public carry may not directly change household gun ownership, but it can change the carrying environment, the perceived normality of immediate firearm availability, and the screening requirements attached to carrying a handgun. Those pathways are closer to lethal-means access and policy screening than to the deterrence theory that dominated earlier right-to-carry debates. For that reason, this study treats firearm suicide as the primary outcome and firearm homicide as a separate secondary contrast.

The main empirical challenge is selection into adoption. States that adopted permitless carry tended to have higher baseline gun ownership, greater rurality, higher Republican presidential vote share, and higher baseline firearm-suicide rates than states that did not adopt. These differences do not preclude panel analysis, but they make interpretation demanding. A post-adoption coefficient can reflect the policy, pre-existing state trajectories, other firearm laws, political selection, health-access differences, substance-use environments, poverty, age structure, or mental-health workforce access.

Legal coding is a second source of uncertainty. Permitless open carry is not the same as permitless concealed carry, and several states do not fit cleanly into a binary adoption map. Vermont entered the panel already permitless, so it is not a within-panel adoption event. Arkansas has competing legal interpretations before and after its 2023 statutory clarification. Mississippi’s 2015 and 2016 changes differ in how broadly permitless concealed carry applied. Ignoring these distinctions can produce a precise estimate attached to an imprecise treatment definition.

We address these concerns in two ways. First, we use a source-audited legal treatment file that distinguishes clean within-panel adopters, verified non-adopters, baseline-permitless states, partial cases, and ambiguous reviewed cases. The audit also records whether adoption removed training requirements, carry-permit background checks, or permit-specific misdemeanor screens. Second, we prespecify an interpretive structure: firearm suicide is the primary outcome; total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks; and mechanism-oriented subgroup analyses are exploratory. State-specific trends, staggered-adoption estimators, Arkansas recoding, leave-one-out estimates, placebo timing, and covariate-layer sensitivity models are used to characterize robustness and limits rather than to create multiple independent headline findings.

The resulting claim is deliberately narrow. This is a source-audited state-panel analysis of where

the permitless-carry association appears, where it persists, and where it weakens. The central pattern is a firearm-suicide association, with total suicide often moving in the same direction and firearm homicide showing no comparable positive signal.

Data and Treatment Coding

State-year panel

The analysis uses an annual 50-state panel from 1999 through 2024. Mortality outcomes come from CDC WONDER underlying-cause mortality files and are expressed as crude rates per 100,000 residents [12]. Crude rates are the primary outcome scale because they measure the realized mortality burden in each state-year. Age-adjusted WONDER rates, standardized to the 2000 U.S. standard population, are used as demographic-composition sensitivity checks for firearm suicide, total suicide, firearm homicide, and total firearm deaths. CDC WISQARS documentation provides the broader injury-surveillance context for fatal injury data derived from National Center for Health Statistics mortality files [13]. The primary clean-adoption models use 1,222 state-year observations from 26 source-verified within-panel adopters and 21 verified non-adopters. Firearm-homicide models use 1,142 observations because some state-year homicide rates are unavailable or suppressed. The primary outcome is firearm suicide. Secondary outcomes are total suicide, non-firearm suicide, firearm homicide, and total firearm deaths.

The baseline state-year covariates are unemployment and per-capita income. Sensitivity layers add external firearm-law indicators from the Tufts State Firearm Law Database [14], health-insurance access from Census SAHIE, drug-overdose mortality, Census demographic and poverty measures, and HRSA mental-health-provider access. RAND state-level household firearm-ownership estimates, rurality, presidential vote share, and baseline firearm-suicide rates are used for descriptive selection and heterogeneity analyses [15].

Treatment coding

The treatment is statewide permitless concealed carry: eligible adults can carry a concealed handgun in public without first obtaining a carry permit. The annual treatment indicator turns on in the first calendar year the statewide permitless concealed-carry rule is effective. Permitless open carry alone is not treated as permitless concealed carry.

The legal audit contains one row per state. It records 26 source-verified current-adopter rows, 21 verified non-adopter rows, one partial row, one ambiguous reviewed row, and one baseline-permitless row (Table 1). The primary analysis excludes the partial, ambiguous, and baseline-permitless rows so the headline estimate compares clean within-panel adopters with verified non-adopters. Mississippi, Arkansas, and Vermont remain visible in the legal audit rather than being hidden inside the binary treatment variable. Arkansas is tested separately under 2021 and 2023 coding, Mississippi is reintroduced in edge-case sensitivity runs, and Vermont is recorded as baseline permitless rather than as a within-panel adoption event. The full state-by-state treatment table in Supplementary Note 2 reports the state, adoption year, audit status, primary-map inclusion flag, coding note, and source URL trail.

Table 1: **Source-audited permitless-carry treatment map.**

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

The primary analytic panel contains 26 clean source-verified adoption-year states and 21 verified non-adopter states. Mississippi, Arkansas, and Vermont are handled outside the primary clean-adoption estimate.

The audit also records carry-permit features for the 26 source-verified adopter rows. Twenty-one had a training requirement removed, 25 removed a carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These fields describe the carry-permit process, not every underlying firearm-possession prohibition. We therefore treat them as descriptive policy features rather than as definitive mechanism tests.

Empirical Strategy

The central specification is a two-way fixed-effects state-year regression:

$$Y_{st} = \alpha_s + \lambda_t + \beta D_{st} + X'_{st}\theta + \varepsilon_{st},$$

Here Y_{st} is the mortality rate for state s in year t , D_{st} equals one in and after the first calendar year statewide permitless concealed carry becomes effective, α_s and λ_t are state and year fixed effects, and X_{st} contains unemployment and per-capita income in the baseline model. Standard errors are clustered by state. The coefficient β is the adjusted post-adoption association: the average within-state change after adoption, net of national year shocks and the baseline covariates. The unweighted model targets an average-state association; population-weighted models are reported separately because they answer the resident-weighted version of the same question.

This estimand is familiar in firearm-policy panel studies, but staggered adoption complicates interpretation when effects vary across cohorts or time [16–19]. We therefore report cohort-window ATT summaries, not-yet-treated dynamic ATT estimates, and never-treated-control event-time fixed-effect estimates. These analyses ask whether the direction of the result is similar under alternative timing comparisons and whether pre-adoption event-time patterns undermine a parallel-trends interpretation [20].

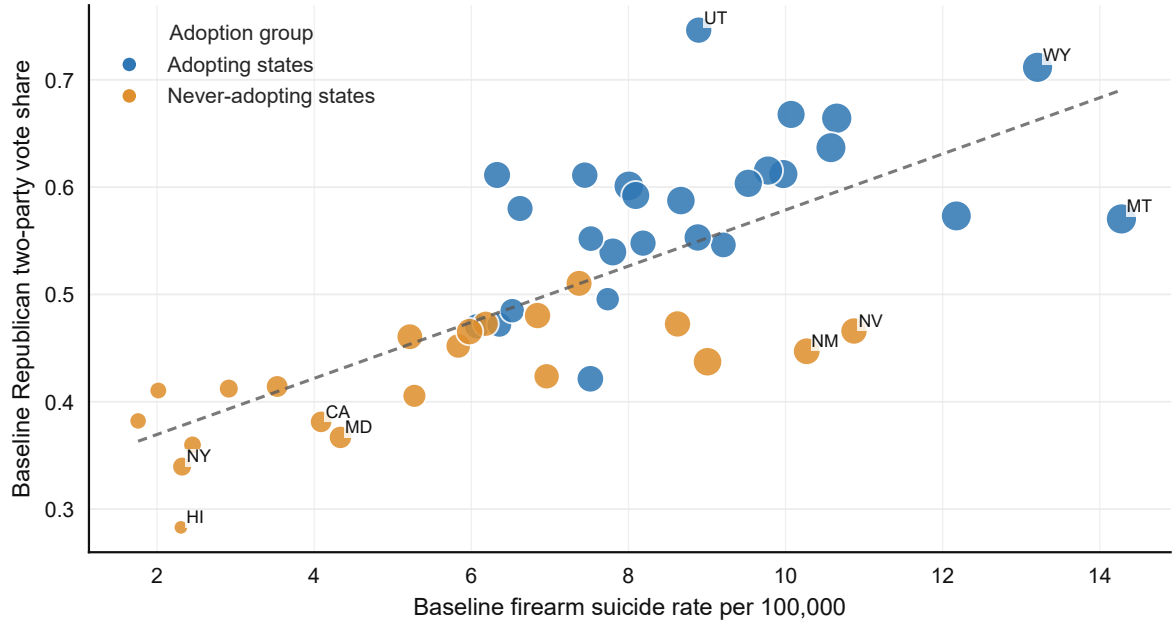
The outcome ordering was fixed before drafting. Firearm suicide is primary. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths assess outcome specificity. Sensitivity analyses add firearm-law covariates, non-firearm contextual covariates, demographic and mental-health covariates, age-adjusted outcome rates, state-specific trends, COVID-period exclusions, Arkansas recoding, leave-one-adopter-out checks, and placebo timing among never-treated states. We do not treat every covariate layer as a superior causal model because some time-varying covariates may reflect broader state policy environments. Heterogeneity and policy-feature analyses are hypothesis-generating because subgroup and feature-specific adopter counts are small.

Results

Adopting states differed before adoption

Before estimating mortality models, we compared clean-adopter and verified non-adopter states on baseline state characteristics (Fig. 1). Adopter states had higher Republican two-party presidential vote share (0.579 vs. 0.421), higher estimated household gun ownership (0.476 vs. 0.322), greater rurality (0.679 vs. 0.433), and higher baseline firearm-suicide rates (8.849 vs. 5.435 deaths per 100,000). These differences make an unadjusted adopter-versus-non-adopter comparison uninformative.

Policy adoption is patterned by baseline risk and politics



Point size is proportional to baseline household firearm ownership. State-level baseline measures describe selection, not treatment effects.

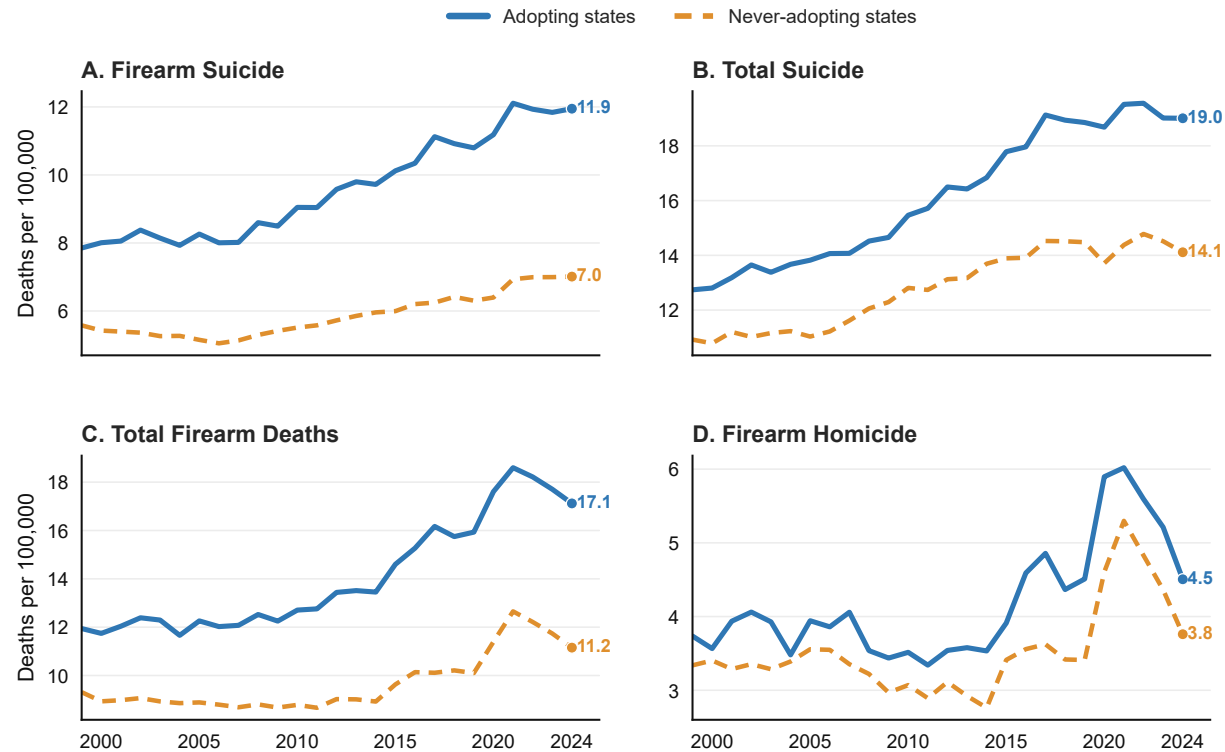
Figure 1: **Political and structural selection into permitless carry adoption.** The descriptive state-level comparison shows that adopter states differed from non-adopter states before or around adoption on political vote share, estimated gun ownership, rurality, and baseline firearm-suicide rates. The selection pattern motivates fixed effects, covariate-layer sensitivity checks, and staggered-adoption diagnostics.

The formal adoption logit was not stable because the small state sample and strong political sorting produced perfect-separation and convergence problems. We therefore use the descriptive comparisons as evidence of selection rather than interpreting unstable logit coefficients. This pattern frames the central empirical limitation: adopting and non-adopting states are not exchangeable without modeling assumptions.

Descriptive trends differed by outcome

Descriptive trends separated suicide and homicide before regression adjustment (Fig. 2). Firearm-suicide and total-suicide rates were higher in adopter states, while homicide trends were noisier and did not visually track the same post-adoption pattern. These trends are not causal evidence because states differed before adoption, but they motivate treating firearm suicide and firearm homicide as distinct endpoints.

Outcome trends by permitless carry adoption status



Annual state means, 1999-2024. Groups are defined by whether a state ever adopts permitless carry during the panel.

Figure 2: **Outcome trends by permitless-carry adoption status.** Descriptive trends motivate separate suicide and homicide analyses. They are not interpreted as causal estimates because adopter and non-adopter states differ structurally and politically before adoption.

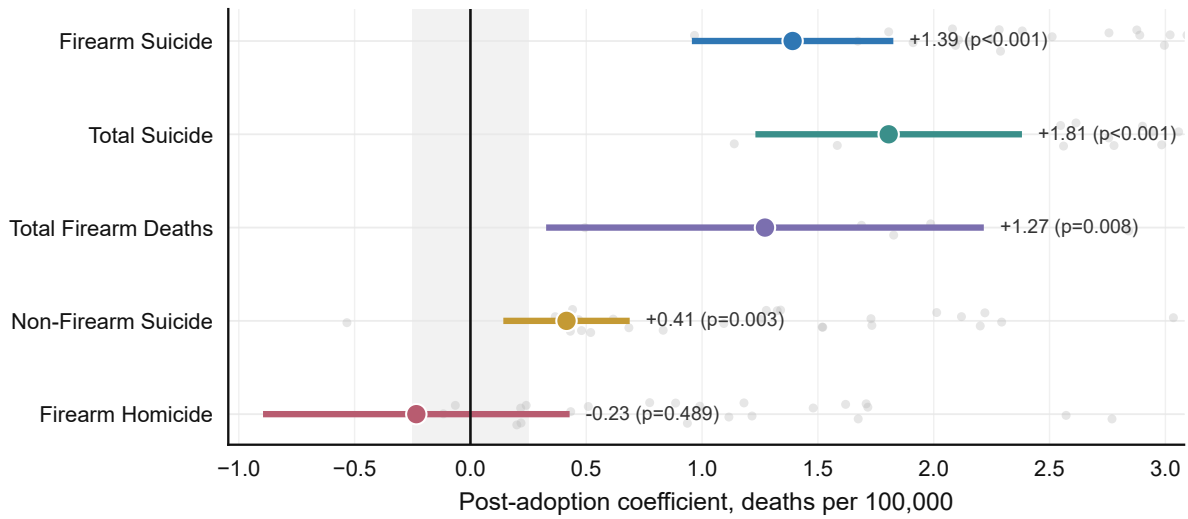
Baseline models showed a suicide association

The baseline fixed-effects model estimated a positive post-adoption association for firearm suicide (Table 2; Fig. 3). Permitless carry adoption was associated with a 1.391 deaths per 100,000 increase in firearm suicide rates (SE 0.222; $p < 0.001$). Total suicide also increased by 1.805 deaths per 100,000 (SE 0.294; $p < 0.001$), and total firearm deaths increased by 1.271 deaths per 100,000 (SE 0.482; $p = 0.008$). Non-firearm suicide was positive but smaller (0.415; SE 0.139; $p = 0.003$).

Table 2: **Baseline fixed-effects mortality estimates.**

Outcome	Coef.	SE	<i>p</i>	N
Firearm suicide	1.391	0.222	< 0.001	1222
Non-firearm suicide	0.415	0.139	0.003	1222
Total suicide	1.805	0.294	< 0.001	1222
Firearm homicide	-0.234	0.338	0.489	1142
Total firearm deaths	1.271	0.482	0.008	1222

Adjusted difference-in-differences estimates



Colored estimates are adjusted TWFE coefficients with +/- 1.96 SE intervals; pale points show unadjusted adopter-state pre/post changes.

Figure 3: **Baseline fixed-effects coefficient forest.** Models include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Firearm suicide, total suicide, and total firearm deaths are positive in the baseline specification; firearm homicide is not.

Age-adjusted outcome rates preserved the main pattern

Because suicide mortality varies strongly by age and adopting states may differ in age composition, we repeated the baseline fixed-effects models using age-adjusted WONDER rates where available (Table 3). These models ask whether the post-adoption association remains after recalculating rates as if states shared the same standard age distribution. Firearm suicide remained positive and similar to the crude-rate estimate (1.407 vs. 1.391 deaths per 100,000), total suicide remained positive, and total firearm deaths remained positive. Firearm homicide remained statistically weak and centered below zero. The age-adjusted homicide model used fewer observations because WONDER marks more small-cell homicide rates as unreliable.

Table 3: **Age-adjusted outcome-rate sensitivity.**

Outcome	Coef.	SE	p	N
Firearm suicide, age-adjusted	1.407	0.228	< 0.001	1220
Total suicide, age-adjusted	1.914	0.327	< 0.001	1222
Total firearm deaths, age-adjusted	1.230	0.510	0.016	1222
Firearm homicide, age-adjusted	-0.230	0.426	0.590	1037

Table 4 places the firearm-suicide estimate beside the main sensitivity checks. The association is positive across many specifications, including population weighting, pre-COVID restriction, COVID-period exclusion, and external firearm-law covariates. The most important attenuation occurs with state-specific linear trends, where the estimate falls to 0.249 ($p = 0.079$). Event-time fixed effects also show a positive post-adoption mean but a statistically significant pre-adoption signal. These results support the direction of the baseline association while limiting a strong causal interpretation.

Table 4: **Firearm-suicide diagnostic spine.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.391	< 0.001	Clean-adoption state-level association
Population-weighted fixed effects	0.919	< 0.001	Resident-weighted association
Pre-COVID only	1.298	< 0.001	Drops post-2019 years
COVID-excluded	1.356	< 0.001	Drops 2020–2021
External firearm-law covariates	1.096	< 0.001	Policy-environment sensitivity
Leave-one-adopter-out	1.265–1.481	–	No sign changes across adopter exclusions
Never-treated placebo timing	1.391 > 0.338	–	Observed estimate exceeds placebo timing distribution
State-specific linear trends	0.249	0.079	Central attenuation caveat
Cohort ATT windows	0.601/0.577/0.641	–	2-, 3-, and 5-year windows
Not-yet-treated dynamic ATT	0.387	–	Positive post-adoption mean
Event-time fixed effects	0.651	0.0000001	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Modern staggered-adoption rows are reported as diagnostics because event-time checks flag pre-adoption signal; the event-time p column reports the minimum pre-adoption p rather than a post-adoption effect test.

192 Firearm-homicide missingness was concentrated in small states

193 Firearm homicide did not follow the same pattern. The baseline fixed-effects estimate was -0.234
194 deaths per 100,000 (SE 0.338; $p = 0.489$), with 1,142 state-year observations. The estimate was
195 statistically weak and centered near zero. Missingness remains relevant: 80 of 1,222 possible clean-
196 primary state-years were missing or suppressed, all in the smallest-population quartile (Table 5).
197 This limitation tempers any strong homicide-null claim, but it does not account for the suicide-
198 homicide contrast because total suicide and total firearm deaths show positive associations where
199 firearm homicide does not.

Table 5: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1222	1142	80	6.5%
Adopter-state state-years	676	610	66	9.8%
Non-adopter-state state-years	546	532	14	2.6%
Q1 smallest	312	232	80	25.6%
Q2	312	312	0	0.0%
Q3	286	286	0	0.0%
Q4 largest	312	312	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

Together, the baseline estimates and missingness audit support an outcome-specific interpretation rather than a broad firearm-violence claim.

Change-score windows showed the same outcome contrast

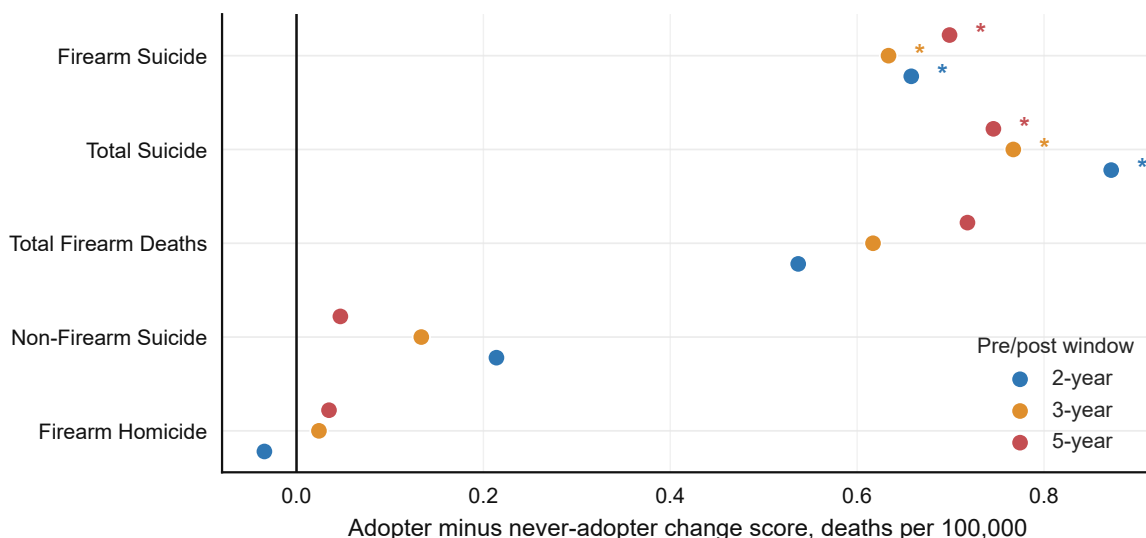
The pre-post change-score design asks a simpler question: within fixed windows around adoption, did adopter states experience larger changes than never-adopter states? Firearm suicide was positive and statistically significant in all three windows. The adopter-minus-never difference was 0.658 deaths per 100,000 in the 2-year window ($p = 0.001$), 0.634 in the 3-year window ($p < 0.001$), and 0.699 in the 5-year window ($p < 0.001$).

Firearm homicide again showed little evidence of a post-adoption increase (Table 6; Fig. 4). The corresponding homicide differences were -0.034 ($p = 0.881$), 0.024 ($p = 0.928$), and 0.035 ($p = 0.899$). Total suicide was positive in all three windows, while total firearm deaths was positive but did not reach $p < 0.05$ in these window checks. Although less adjusted than the fixed-effects model, the change-score comparison shows that the suicide-homicide contrast is not unique to the baseline regression specification.

Table 6: **Pre-post change-score window checks.**

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.658**	0.634***	0.699***
Non-firearm suicide	0.214	0.134	0.047
Total suicide	0.872**	0.767***	0.746**
Firearm homicide	-0.034	0.024	0.035
Total firearm deaths	0.537	0.617	0.718

Change-score comparisons across robustness windows



Positive values indicate larger pre/post increases in adopting states. Asterisks denote Welch-test $p < 0.05$.

Figure 4: **Change-score robustness across pre-post windows.** Firearm suicide is positive across 2-, 3-, and 5-year windows. Firearm homicide remains statistically weak across the same windows.

Covariate layers attenuated the firearm-suicide estimate

Permitless carry may move alongside other firearm policies. We therefore added covariates for permit-to-purchase laws, waiting periods, universal background checks, ERPO/red-flag laws, safe-storage laws, stand-your-ground laws, and dealer licensing using the Tufts database [14]. The firearm-suicide estimate decreased from 1.391 to 1.096 deaths per 100,000 but remained statistically significant ($p < 0.001$). Total suicide and total firearm deaths also remained positive; firearm homicide remained statistically weak.

Suicide-related social environments may also differ across adopting and non-adopting states. Health-access covariates reduced the firearm-suicide estimate to 0.707 ($p < 0.001$), overdose covariates reduced it to 0.492 ($p = 0.060$), and demographic-poverty covariates reduced it to 0.695 ($p < 0.001$). These results suggest that the firearm-suicide association is not explained entirely by these measured domains, while the short overdose window marks a boundary for statistical precision.

The strongest attenuation came from specifications that included mental-health-provider access (Table 7). The HRSA mental-health-provider window is short, leaving only 141 state-year observations for the mental-health and full Phase 3B2 models. In that shorter window, firearm suicide remained positive but was no longer statistically significant (0.116, $p = 0.744$ for mental-health access; 0.176, $p = 0.633$ for the combined Phase 3B2 specification). These rows should not replace the full-panel model, but they identify an important boundary: the association weakens in a short recent window after adding mental-health-provider access and the full Phase 3B2 covariate set.

These models are best interpreted as sensitivity analyses, not as proof that adding more time-varying covariates necessarily improves causal identification. Some covariates, including overdose mortality, health access, poverty, and mental-health-provider access, may themselves reflect broader state policy environments or shocks related to adoption.

Table 7: **Firearm-suicide estimate across covariate-layer sensitivities.**

Specification	Coef.	<i>p</i>	N
Baseline fixed effects	1.391	< 0.001	1222
External firearm-law covariates	1.096	< 0.001	1222
Health-access covariates	0.707	< 0.001	752
Overdose covariates	0.492	0.060	282
Demographic-poverty covariates	0.695	< 0.001	940
Mental-health access covariates	0.116	0.744	141
Full Phase 3B2 covariates	0.176	0.633	141

Staggered-adoption analyses raised pretrend concerns

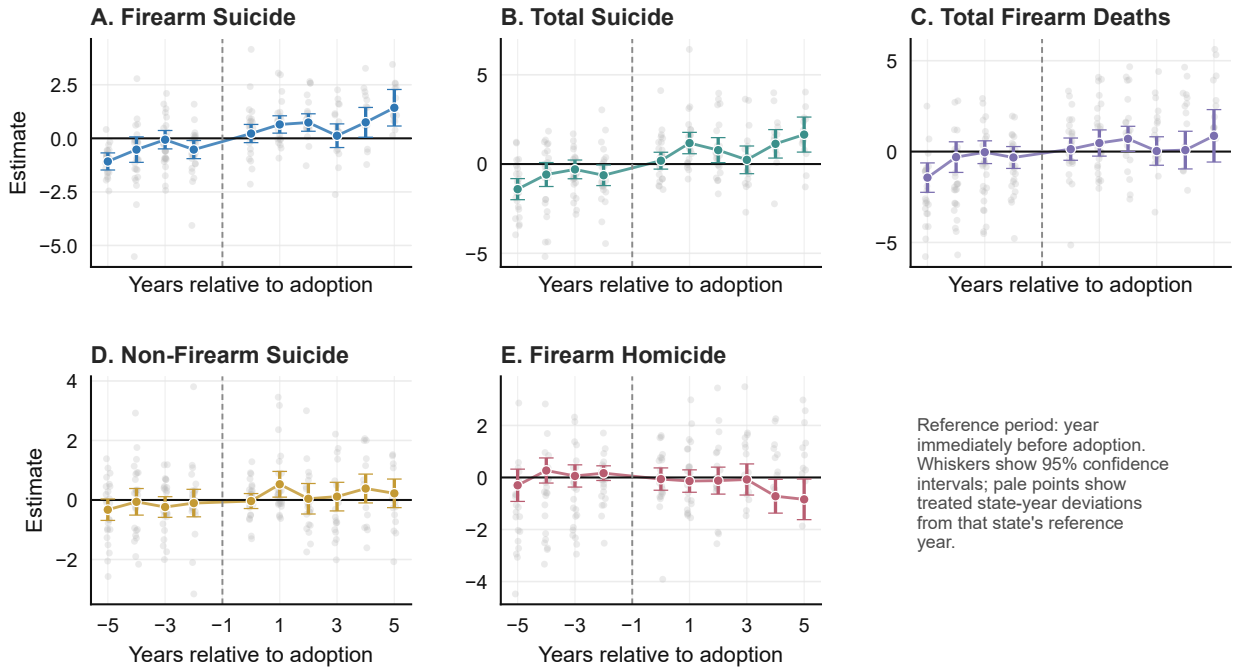
Because adoption is staggered across states, we next examined cohort-window and event-time estimates (Table 8). Cohort-window ATT estimates for firearm suicide were positive across the 2-, 3-, and 5-year windows: 0.601, 0.577, and 0.641 deaths per 100,000. The not-yet-treated dynamic ATT summary also had a positive post-adoption mean for firearm suicide (0.387), while firearm homicide had a negative post-adoption mean (-0.157).

Table 8: **Staggered-adoption diagnostic summary.**

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.601	0.577	0.641	0.387	Positive, pretrend flag
Non-firearm suicide	0.270	0.239	0.241	0.077	Positive diagnostic
Total suicide	0.871	0.816	0.882	0.464	Positive, pretrend flag
Firearm homicide	-0.004	-0.107	-0.106	-0.157	No positive diagnostic
Total firearm deaths	0.382	0.435	0.493	0.200	Positive, pretrend flag

The event-time estimates are more cautionary (Fig. 5). Firearm suicide had positive post-adoption coefficients, but it also had statistically significant pre-adoption coefficients in the never-treated-control event-time model. Total suicide and total firearm deaths showed similar pretrend patterns. Firearm homicide had no positive post-adoption pattern. These results are consistent with the direction of the suicide association but do not support a clean parallel-trends interpretation.

Event-study estimates around adoption



Dynamic coefficients are plotted relative to the year before permitless carry adoption; background points show the underlying treated-state event-time data.

Figure 5: **Event-time diagnostic estimates.** Event-time models show post-adoption suicide-related increases but also pre-adoption signals for firearm suicide, total suicide, and total firearm deaths. Those pretrend flags prevent the staggered-adoption results from serving as confirmatory causal estimates.

Robustness checks identified trend sensitivity

The firearm-suicide estimate remained positive under COVID-period exclusion (1.356; $p < 0.001$), pre-COVID restriction (1.298; $p < 0.001$), population weighting (0.919; $p < 0.001$), and external firearm-law covariates (1.096; $p < 0.001$). Leave-one-adopter-out estimates ranged from 1.265 to 1.481, and the observed firearm-suicide coefficient exceeded the 95th percentile of the never-treated placebo-timing distribution.

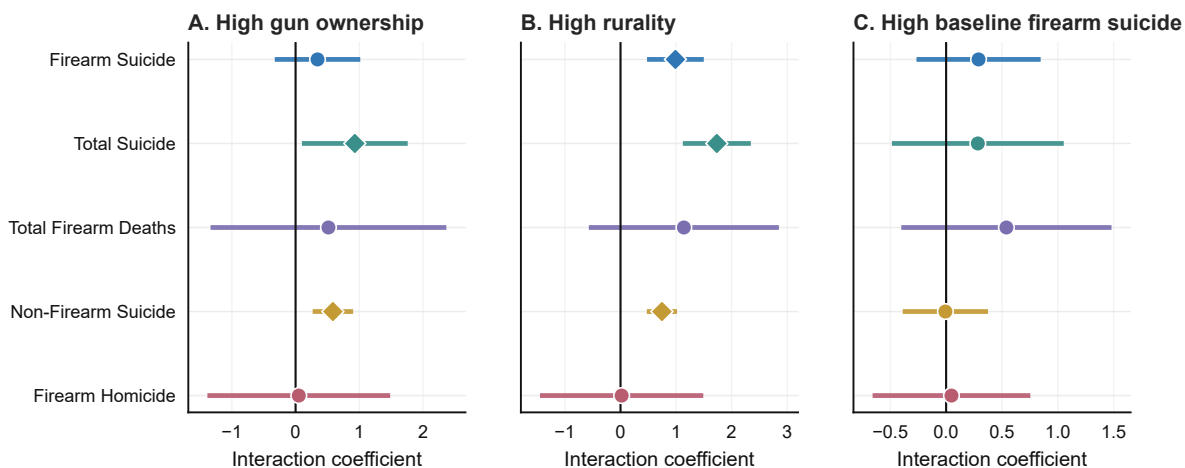
State-specific linear trends were the main exception. With state trends, the firearm-suicide estimate attenuated to 0.249 and was not statistically significant at the 0.05 level ($p = 0.079$). Total firearm deaths also attenuated strongly (0.070; $p = 0.806$). Total suicide remained positive (0.372; $p = 0.025$). These rows show that part of the firearm-suicide association is difficult to distinguish from longer state-specific trajectories.

Legal edge-case sensitivity did not change the broad pattern. When Arkansas was coded as treated in 2021 or 2023, firearm suicide remained positive and significant (1.361 and 1.379), total suicide remained positive and significant (1.777 and 1.785), and firearm homicide remained statistically weak (-0.205 and -0.250). Reintroducing Mississippi alone produced a firearm-suicide estimate of 1.300, and the combined Mississippi-plus-Arkansas scenarios remained positive and significant. The main outcome contrast is therefore not driven by these edge-case coding decisions.

Heterogeneity patterns were strongest for rurality

The clearest heterogeneity pattern was rurality (Fig. 6). Firearm suicide had a positive high-rurality interaction (0.988; $p < 0.001$), total suicide also had a positive high-rurality interaction (1.733; $p < 0.001$), and non-firearm suicide had positive interactions for high rurality (0.745; $p < 0.001$) and high estimated gun ownership (0.586; $p < 0.001$). Total firearm-death and firearm-homicide heterogeneity estimates were not statistically strong.

Heterogeneity in post-adoption associations



Interaction terms estimate whether associations are larger in states above each baseline median; diamonds mark $p < 0.05$.

Figure 6: **Exploratory heterogeneity checks.** Rurality is the clearest recurring heterogeneity dimension for firearm suicide, total suicide, and total firearm deaths. These models are exploratory because subgroup indicators are coarse and adoption is politically patterned.

The data do not directly observe carrying frequency, firearm acquisition, firearm storage, crisis timing, impulsive crises, or household firearm access. Plausible pathways include reduced pre-carry screening, reduced delay, weaker training requirements, and normalization of firearm availability, but the state-year panel cannot distinguish among them. The mortality pattern is therefore consistent with higher suicide-related risk after adoption, but it does not identify a specific firearm-access mechanism.

The policy-feature models did not identify a stable mechanism. None of the 15 training, permit-background-check, or misdemeanor-screen interaction rows had $p < 0.05$. The permit-background-check interaction could not be estimated because all 26 source-verified adopter rows removed the carry-permit background-check screen or an equivalent carry-permit screen, leaving no source-verified comparison group. Training removal and misdemeanor-screen removal had estimable rows, but the interaction estimates were not statistically strong. These results are best read as descriptive feature checks rather than evidence that one permit component explains the mortality association.

Discussion

This study evaluates a specific, recent form of concealed-carry deregulation: the removal of the carry-permit requirement. It does not estimate the full history of right-to-carry laws, and it does

not test whether public carrying deters crime. In this 50-state panel, the clearest pattern was outcome-specific: permitless carry adoption was more consistently associated with suicide-related mortality than with firearm homicide.

The firearm-suicide estimate was positive in the baseline fixed-effects model, age-adjusted outcome models, change-score windows, population-weighted models, Arkansas recodings, leave-one-adopter-out checks, and several covariate and staggered-adoption diagnostics. Total suicide and total firearm deaths often moved in the same direction. Firearm homicide did not. This contrast is the central empirical result.

The pattern is not a clean method-substitution result. If permitless carry only shifted suicide method toward firearms, firearm suicide would rise while total suicide remained stable or non-firearm suicide fell. Instead, total suicide increased in the baseline fixed-effects model, and non-firearm suicide was positive in that same specification. The non-firearm and total-suicide results were less stable across sensitivity layers, so firearm suicide remains the most durable signal. Still, the broader suicide pattern argues against describing the finding as simple substitution across methods.

The main limitation is identification. State-specific linear trends attenuated the firearm-suicide estimate, and event-time models showed pre-adoption signals for firearm suicide, total suicide, and total firearm deaths. These patterns suggest that adopting states were already on different mortality trajectories, or that policy timing coincided with other changes not fully captured by the covariates. The estimates should therefore be read as associational evidence, not proof that permitless carry caused the observed suicide increases.

The legal audit is an important part of the study because treatment coding is itself a source of uncertainty. Arkansas, Mississippi, and Vermont illustrate the problem. A state can be baseline permitless, partially permitless, ambiguously interpreted, or newly clarified by statute. A state can also remove the pre-carry permit application while retaining underlying possession prohibitions. The audit makes those distinctions visible and prevents the treatment variable from appearing more certain than the law allows.

The feature analysis should also be interpreted cautiously. Training removal, background-check removal, and misdemeanor-screen removal are plausible pathways, and prior concealed-carry work has emphasized the importance of permit provisions [2]. In this dataset, however, feature groups are small or nearly universal among clean adopters. The absence of statistically significant feature interactions is therefore not strong evidence against those pathways; it more likely reflects limited variation in the current state panel.

These findings fit a broader firearm-policy literature that increasingly treats suicide and homicide as distinct outcomes. State firearm laws associated with lower homicide are not always the same as those associated with lower suicide [21, 22]. Recent suicide-specific evidence links handgun permits, waiting periods, and concealed-carry permit requirements to lower firearm suicide rates [7]. The present analysis extends that logic to the recent permitless-carry wave and finds that the clearest mortality association is again on the suicide side.

Several limitations remain beyond identification. The panel is state-level and cannot identify individual behavior, carrying frequency, firearm acquisition, storage, mental-health crises, crisis timing, or interpersonal circumstances. Some homicide observations are missing because of suppression or incomplete availability, with missingness concentrated in small-population state-years. The HRSA mental-health-provider specifications use a short recent window, and the policy-feature fields require deeper statute-level auditing before they can support mechanism claims.

The practical implication is primarily methodological. Permitless-carry studies should separate primary outcomes from secondary and exploratory checks, audit legal treatment timing, report edge-case coding, report both unweighted and population-weighted estimands, and use staggered-adoption diagnostics without treating them as a complete solution to identification problems. Under

that standard, the strongest statement from this analysis is cautious: adopting states experienced larger post-adoption increases in firearm suicide rates, while firearm homicide showed no comparably consistent evidence of change in the available data.

Additional Methods Details

All baseline fixed-effects models used ordinary least squares with state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Robustness specifications excluded 2020–2021, restricted the panel to pre-2020 years, applied population weights, added state-specific linear trends, or added external firearm-law covariates. Covariate-layer sensitivity models added firearm-law domains, uninsured rates, overdose mortality, age structure, race and ethnicity shares, poverty rates, and mental-health-provider access as described in the repository output tables. The unweighted estimates target the average state-level association; the population-weighted estimates target the average resident-level association.

Change-score analyses computed, for each adopter state, the difference between the average post-adoption rate and the average pre-adoption rate in 2-, 3-, and 5-year windows. Never-adopter states were assigned comparable window summaries and compared with Welch tests. Staggered-adoption checks included cohort-window ATT estimates, not-yet-treated dynamic ATT estimates, and never-treated-control event-time fixed-effect estimates. Age-adjusted outcome models used CDC WONDER age-adjusted rates as alternate outcomes and retained the same state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors as the crude-rate baseline model. These checks were designed as sensitivity analyses because several outcomes showed pre-adoption event-time signals.

The outcome order was fixed before drafting the paper. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Firearm-law covariates, contextual covariate layers, age-adjusted outcome rates, staggered-adoption estimators, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity analyses. Rurality, gun ownership, baseline-risk heterogeneity, and policy-feature interactions are exploratory.

Data and Code Availability

Processed analysis outputs, source-audited policy coding files, figure-generation scripts, and the manuscript-ready result tables are maintained in the project repository. Raw CDC mortality files and firearm-law source files are governed by their originating agencies and are documented in the repository source appendix.

Author Contributions

B.K. and Y.W. assembled the policy audit, implemented the mortality panel analyses, generated figures and tables, interpreted the results, and drafted the manuscript.

Competing Interests

The authors declare no competing interests.

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